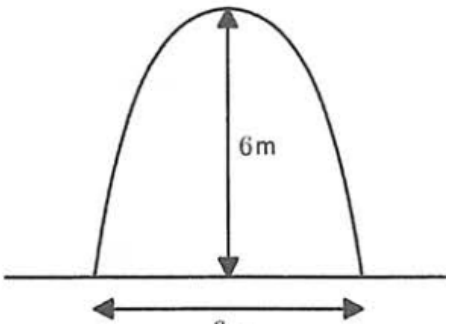
Hyperbolic Functions: Modelling

Questions 1–8

Question 1. The cross-section of a tunnel is modelled by the curve with equation

$$y = 2p - q\cosh x$$

where y is the height, in metres, of the tunnel above the road surface, x is the horizontal distance, in metres, from the centre of the tunnel, and p and q are positive constants. The tunnel is symmetric about the vertical line $x = 0$.



The tunnel is 6 metres wide at road level and has a maximum height of 6 metres.

a) Find the values of p and q to three significant figures.

b) A vehicle of rectangular cross-section is to pass through the tunnel. The vehicle is 4 metres wide and 4 metres high. Determine whether the vehicle can pass through the tunnel without touching the sides.

Prerequisite knowledge for Q2: Mean Value and Reversing the Chain Rule

Question 2. The temperature, $T°\text{C}$, of a chemical solution t hours after a catalyst is added can be modelled by the equation

$$T = 60 - 15\tanh 0.8t$$

for $t \geq 0$. Using the model, determine the mean temperature of the solution during the first 45 minutes of the catalyst being added, giving your answer correct to two decimal places.

Question 3. The velocity, $v\,\mathrm{m\,s^{-1}}$, of a skydiver t seconds after leaving a stationary helicopter is modelled by

$$v = 55\tanh(0.12t)$$

for $t \geq 0$.

a) Explain why this model predicts the existence of a terminal velocity, and state its value.

b) According to the model, find the time taken for the skydiver to reach 90% of the terminal velocity. Give your answer to 2 decimal places.

Prerequisite knowledge for Q4: Volume of Revolution and Hyperbolic Identities

Question 4. A water treatment company is designing a large filtration pipe. The inside wall of the pipe is modelled by the curve

$$y = 2 + \cosh\left(\frac{x}{2}\right)$$

for $-4 \leq x \leq 4$, where x and y are measured in metres. The pipe is formed by rotating the region enclosed by the curve and the x-axis about the x-axis.

a) Show that the volume V of the pipe is 381 m^3 to the nearest cubic metre.

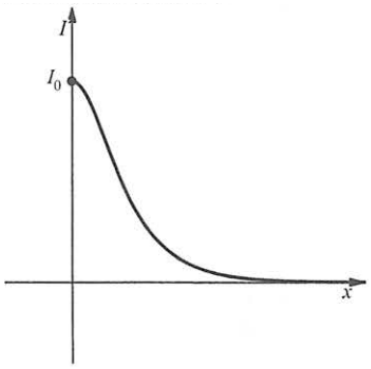
b) The pipe can process 0.85 m^3 of water per second when completely full. Estimate how many seconds are needed to completely fill the pipe.

Prerequisite knowledge for Q5: Differentiation of Hyperbolic Functions

Question 5. Light travelling through water decreases in intensity with depth. In a particular model, the intensity I at a depth of x metres below the surface is given by

$$I = I_0 \operatorname{sech}(kx)$$

where I_0 is the intensity at the surface, k is a positive constant, and $x \geq 0$.



a) Show that the depth x at which the intensity is equal to fI_0, where $0 < f < 1$, is given by $x = \dfrac{1}{k} \operatorname{arcosh}\left(\dfrac{1}{f}\right)$.

b) Given that the intensity is reduced to 40% of its surface value at a depth of 12 metres, find the value of k, giving your answer to 3 significant figures.

c) Given that $I_0 = 120$, find an expression for $\dfrac{dI}{dx}$ in terms of k and x.

d) Using your value of k from part (b), find the rate of change of light intensity with respect to depth when $x = 8$, giving your answer to 3 significant figures.

e) Show that the magnitude of $\dfrac{dI}{dx}$ decreases as x increases for $x \geq 0$. Hence explain what this suggests about the behaviour of light as it enters the water.

f) Given that $\operatorname{sech} u \approx 1 - \frac{u^2}{2}$ for small values of u, comment on how well the model represents light intensity close to the surface.

Question 6.

a) Show that $e^{kt} \equiv \cosh kt + \sinh kt$.

The displacement, x metres, of a particle at time t seconds is governed by the differential equation

$$\frac{d^2x}{dt^2} - 9x = 0.$$

Initially, the particle is at $x = 2$ and is moving with velocity 3 $\mathrm{m\,s^{-1}}$.

b) Show that the general solution can be written in the form $x = A\cosh 3t + B\sinh 3t$, where A and B are constants to be found.

c) Express the solution in the form $x = R\cosh(3t + \alpha)$.

d) Find the time at which the displacement of the particle is $2\sqrt{3}$ metres, giving your answer in the form $\dfrac{1}{6}\ln(a + b\sqrt{3})$, where a and b are rational numbers.

Prerequisite knowledge for Q7: Differentiation and Integration of Hyperbolic Functions

Question 7. A small research rocket is launched vertically upwards. Its velocity, $v\,\mathrm{m\,s^{-1}}$, at time t seconds after launch is modelled by the equation

$$v = A\sinh kt$$

where A and k are positive constants.

a) Show that the acceleration of the rocket, $a\,\mathrm{m\,s^{-2}}$, can be expressed in the form $a = k\sqrt{v^2 + A^2}$.

b) It is given that the initial acceleration of the rocket is $2\ \mathrm{m\,s^{-2}}$. It is also known that when the rocket reaches a velocity of $12\ \mathrm{m\,s^{-1}}$, its acceleration is $5.2\ \mathrm{m\,s^{-2}}$. Find the value of A.

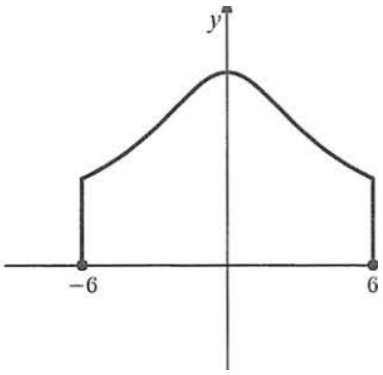
c) Show that $k = 0.4$.

d) Find the distance travelled by the rocket in the first 5 seconds of its launch, giving your answer to the nearest metre.

Prerequisite knowledge for Q8: Integration using Inverse Hyperbolic Functions

Question 8. A farmer is designing a new barn for storing hay. The cross-section of the barn is modelled for $-6 \le x \le 6$ by the curve

$$y = \frac{12}{\sqrt{x^2 + 9}}.$$



The region enclosed by the curve, the x-axis, and the lines $x = -6$ and $x = 6$ represents the cross-sectional area of the barn. Each unit on the axes represents 2 metres. The barn is modelled as a prism of length 18 metres.

a) Show that the cross-sectional area of the barn is $96\ln(2 + \sqrt{5})\,\mathrm{m^2}$.

b) Find the volume of the barn, giving your answer to the nearest $\mathrm{m^3}$.

c) The farmer stores hay bales occupying $1.8\ \mathrm{m^3}$. Estimate the maximum number of hay bales that can fit inside the barn.